

Introduction to Electronics Engineering

MODULE 3 — PART 2: QUESTION BANK

Module 3 (07 hrs): Sinusoidal Oscillators & Operational Amplifiers

For First-Year Students — with complete answers and diagrams
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 – Q23
Section B	7 marks	Compulsory — from every module	Q1 – Q8

- Module 3 has **no 5-mark section** — 5-mark questions are asked only from Modules 1 and 2.
- Every topic of Module 3 has a 2-mark question, and every topic is also covered in the 7-mark section — no topic is left out.
- No question is repeated between sections.
- Answer 2-mark questions in 2–4 lines and 7-mark questions with full explanation and diagrams.
- A numerical example is included in the Wien bridge oscillator topic.

SECTION A — 2-Mark Questions

Q1. What is feedback? What is a feedback amplifier?

Answer: Feedback is a process in which a fraction of the output (voltage or current) is fed back to the input. An amplifier that works on the principle of feedback is called a feedback amplifier.

Q2. Define positive feedback.

Answer: Positive feedback is a process in which a portion of the output signal is fed back to the input **in the same phase**, thereby increasing the net input signal. It is used in oscillators.

Q3. Define negative feedback.

Answer: Negative feedback is a process in which a portion of the output signal is fed back to the input **in opposite phase**, thereby reducing the net input signal. It is used in amplifiers to improve stability.

Q4. Give any two differences between positive and negative feedback.

Answer: Positive feedback: output fed back in phase, gain increases, used in oscillators. Negative feedback: output fed back out of phase, gain decreases, distortion reduces, used in amplifiers.

Q5. What is an oscillator?

Answer: An oscillator is a circuit that produces an AC voltage of the desired frequency and waveshape without any external input signal. It works on positive feedback, satisfying the condition $\beta A = 1$.

Q6. State the Barkhausen criterion for oscillation.

Answer: The Barkhausen criterion states that for sustained oscillation the loop gain must be unity, i.e. **$\beta A = 1$** . If βA is exactly 1, self-sustained oscillations result.

Q7. What is a sinusoidal oscillator? What is a square-wave oscillator?

Answer: If the output signal varies sinusoidally, the circuit is a **sinusoidal oscillator**. If the output voltage rises quickly to one level and later drops quickly to another level, the circuit is a **pulse or square-wave oscillator**.

Q8. What is an op-amp? Why is it so named?

Answer: An operational amplifier is a high-gain voltage amplifier with two inputs and one output that amplifies the difference between the two input voltages. It is so named because it was originally designed to perform mathematical operations (addition, subtraction, integration, etc.) in analog computers.

Q9. Draw the symbol of an op-amp and name its terminals.

Answer: The op-amp is represented by a triangular symbol with two input terminals and one output terminal. The terminal with the negative sign is the inverting input and the terminal with the positive sign is the non-inverting input.

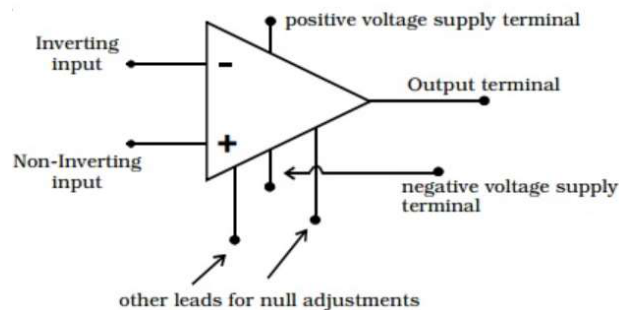


Figure 1: Symbol of the op-amp.

Q10. Write the output equation of an op-amp. What is V_d ?

Answer: $V_o = A_{OL} V_d = A_{OL} (V_1 - V_2)$, where V_d is the difference voltage between the two inputs and A_{OL} is the open-loop gain.

Q11. What is the equivalent circuit of an op-amp?

Answer: The circuit that represents the op-amp parameters in terms of physical components — the input resistance R_{in} , the output resistance R_o and the controlled voltage source $A_{OL} V_d$ — for analysis purposes, is called the equivalent circuit of the op-amp.

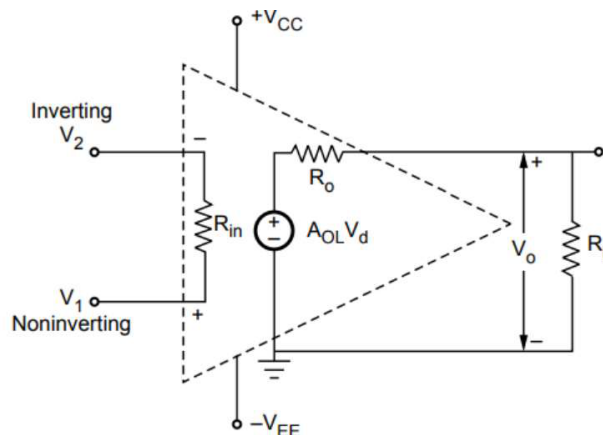


Figure 2: Equivalent circuit of the op-amp.

Q12. List any four characteristics of an ideal op-amp.

Answer: Infinite voltage gain, infinite input impedance (no input current), zero output impedance, and infinite bandwidth. (Also: zero offset voltage, infinite slew rate, infinite CMRR.)

Q13. Define slew rate.

Answer: The slew rate of an op-amp is the maximum rate of change of the output voltage per unit time. It is expressed in volts per microsecond ($V/\mu s$).

Q14. Define CMRR and common-mode gain.

Answer: CMRR (common-mode rejection ratio) is the ratio of the differential gain to the common-mode gain: $CMRR = A_d/A_{cm}$. Common-mode gain is the gain when the same signal is applied to both inputs.

Q15. Compare an ideal op-amp with a practical op-amp (any three points).

Answer: Ideal: infinite gain, infinite input impedance, zero output impedance. Practical (741): very high but finite gain (about 100000), high but finite input impedance (about 2 M-ohm), and low but non-zero output impedance (about 75 ohm).

Q16. Name any four applications of op-amps.

Answer: Signal amplification, active filters (low-pass, high-pass, band-pass), mathematical operations (integration, differentiation, summing), voltage follower (buffer), oscillators and signal generators, and analog computation.

Q17. What is a virtual ground?

Answer: A virtual ground is the condition of the inverting input node of an op-amp, which sits at nearly 0 V (the same potential as ground) without being physically connected to ground, and through which no current flows.

Q18. Write the gain of an inverting amplifier.

Answer: $A_v = -R_f / R_1$, where R_f is the feedback resistor and R_1 is the input resistor. The negative sign indicates 180° phase inversion.

Q19. Write the gain of a non-inverting amplifier.

Answer: $A_{CL} = 1 + R_f / R_1$. The gain is positive, so the output is in phase with the input.

Q20. What is a voltage follower? When does a non-inverting amplifier act as one?

Answer: A voltage follower (buffer) is a circuit with a gain of exactly 1 that provides high input impedance and low output impedance. A non-inverting amplifier acts as a voltage follower when $R_f = 0$ and R_1 is infinite.

Q21. What is a summing amplifier?

Answer: A summing amplifier is an inverting-amplifier-based circuit that adds several input voltages, each multiplied by a constant gain. Its output is $V_o = -R_f(V_1/R_1 + V_2/R_2 + V_3/R_3)$.

Q22. What is an RC phase-shift oscillator? Why is the RC network used?

Answer: It is a sinusoidal oscillator that uses an amplifier and a three-section RC network. The RC network provides the 180° phase shift needed to make the feedback positive, which is necessary for oscillation.

Q23. State the frequency of oscillation of a Wien bridge oscillator.

Answer: $f_0 = 1 / (2 \pi R C)$, when $R_1 = R_2 = R$ and $C_1 = C_2 = C$. The amplifier gain must be 3 (i.e. $R_3/R_4 = 2$).

SECTION B — 7-Mark Questions

Q1. Explain the concept of feedback in amplifiers in detail with a block diagram. Compare positive and negative feedback.

Answer:

The purpose of an amplifier is to amplify the input signal without changing its characteristics except its amplitude. When a fraction of the output is returned to the input, the amplifier is called a feedback amplifier. The input V_s is combined with the feedback signal V_f in a mixer to give $V_i = V_s \pm V_f$, which drives the amplifier.

Positive feedback feeds the output back in phase, increasing the net input and the gain; it is used in oscillators. **Negative feedback** feeds the output back in opposite phase, reducing the gain but improving stability and reducing distortion; it is used in amplifiers.

Parameter	Negative Feedback	Positive Feedback
Phase relationship	Out of phase	In phase
Gain	Reduced	Increased
Stability	Increases	Decreases
Distortion	Reduces	Increases
Used in	Amplifiers	Oscillators

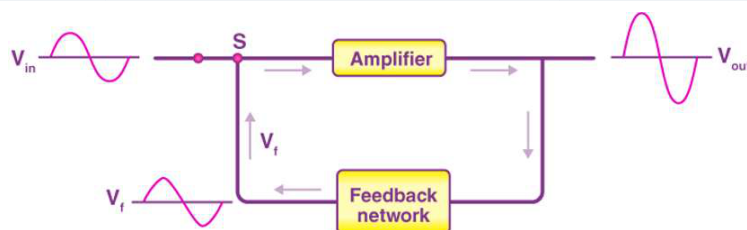


Figure 3: Block diagram of a feedback amplifier.

Block diagram of a feedback amplifier (positive / negative feedback)

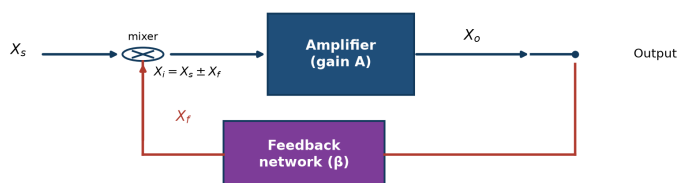


Figure 4: Signal flow in the feedback amplifier.

Q2. State and explain the Barkhausen criterion. How does an oscillator start and sustain its oscillations? Explain with waveforms.

Answer:

The Barkhausen criterion states that for sustained oscillation the loop gain of the amplifier with positive feedback must be exactly unity: **beta A = 1**. No external input is required — only this condition must be satisfied.

- If $\beta A < 1$, the oscillations are damped and die out.
- If $\beta A = 1$, the oscillations are sustained at constant amplitude.
- If $\beta A > 1$, the oscillations grow in amplitude until limiting brings them to a steady state.

In practice, a circuit with $\beta A > 1$ starts oscillating by amplifying the tiny noise voltage always present in the circuit; the waveform is never exactly sinusoidal. The closer βA is to 1, the more nearly sinusoidal the output becomes.

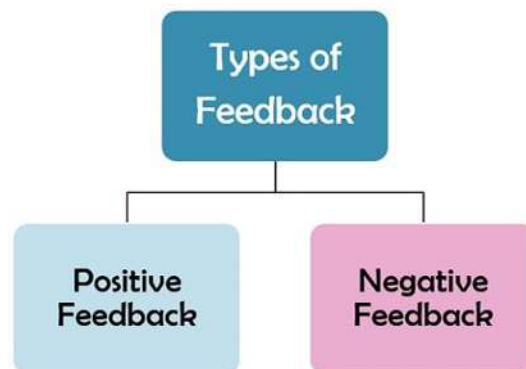


Figure 5: Oscillation conditions for different βA values.

- $|A\beta| < 1 \rightarrow$ Oscillations die out
- $|A\beta| > 1 \rightarrow$ Amplitude grows (distortion)
- $|A\beta| = 1 \rightarrow$ Sustained oscillations

Figure 6: Damped oscillation ($\beta A < 1$).

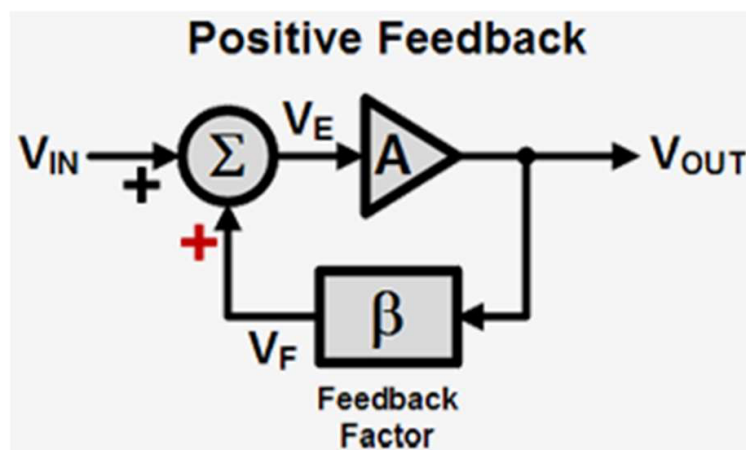


Figure 7: Sustained oscillation ($\beta A = 1$).

1. Magnitude Condition

$$|A\beta| = 1$$

2. Phase Condition

$$\angle A\beta = 0^\circ \text{ or } 360^\circ$$

Figure 8: Growing oscillation ($\beta A > 1$).

Q3. Explain the characteristics of an ideal op-amp in detail. Also give the equivalent circuit and compare the ideal and practical op-amp.

Answer:

- **Infinite voltage gain:** the differential open-loop gain A is infinite.
- **Infinite input impedance:** no current flows into either input, $I_1 = I_2 = 0$.
- **Zero output impedance:** the output voltage is unaffected by the load.
- **Zero offset voltage:** zero output for zero input.
- **Infinite bandwidth:** satisfactory performance over all frequencies.
- **Infinite slew rate:** the output changes simultaneously with the input.
- **Infinite CMRR:** complete rejection of common-mode noise, $CMRR = A_d/A_{cm}$.

Op-amp symbol: the op-amp is drawn as a triangle with two input terminals — the inverting (–) input and the non-inverting (+) input — and one output terminal. It amplifies only the difference of the two inputs, $V_d = V_1 - V_2$.

The equivalent circuit represents the op-amp by its input resistance R_{in} , output resistance R_o and the controlled source $A_{OL}V_d$. The op-amp amplifies only the difference voltage, so the output polarity is decided by the polarity of V_d .

Parameter	Ideal Op-Amp	Practical Op-Amp (741)
Open-loop gain	Infinite	Very high (about 100000)
Input impedance	Infinite	High (about 2 M-ohm)
Output impedance	Zero	Low (about 75 ohm)
Bandwidth	Infinite	Limited (about 1 MHz)

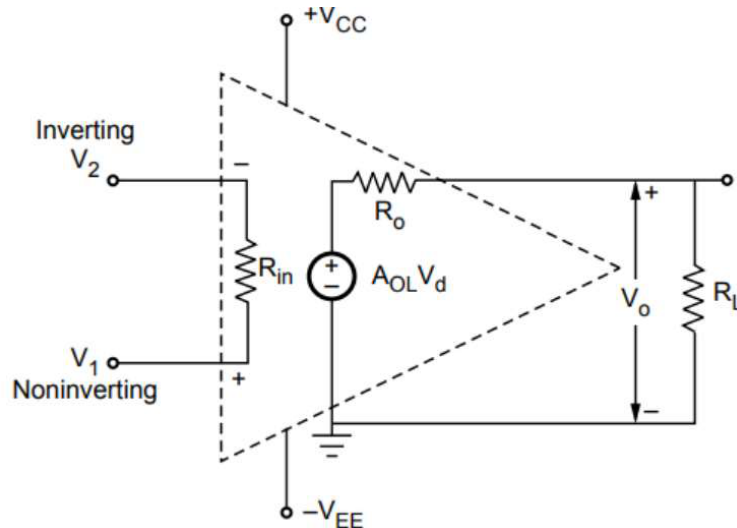


Figure 9: Equivalent circuit of the op-amp.

Property	Ideal	Practical(Typical)
Open-loop gain	Infinite	Very high (>10000)
Open-loop bandwidth	Infinite	Dominant pole(≈ 10 Hz)
CMRR	Infinite	High (> 60 dB)
Input Resistance	Infinite	High (>1 M Ω)
Output Resistance	Zero	Low(< 100 Ω)
Input Bias Currents	Zero	Low (< 50 nA)
Offset Voltages	Zero	Low (< 10 mV)
Offset Currents	Zero	Low (< 50 nA)
Slew Rate	Infinite	A few V/ μ s

Figure 10: Ideal versus practical op-amp parameters.

Q4. Explain the inverting amplifier in detail (circuit, virtual ground, gain derivation, characteristics and applications).

Answer:

Circuit: the input V_1 is applied through R_1 to the inverting input, the output is fed back through R_f to the same input, and the non-inverting input is grounded.

Virtual ground: the inverting terminal is not physically connected to ground but behaves like ground (0 V) because of the virtual short; no current flows into the op-amp input.

Gain derivation:

$$I_1 = V_1/R_1, I_f = -V_o/R_f; \text{ since } I_1 = I_f:$$

$$A_v = V_o/V_1 = -R_f/R_1$$

Characteristics: 180° phase inversion, gain = $-R_f/R_1$, high input impedance with feedback, and a virtual ground at the inverting node.

Applications: summing amplifier, analog integration and differentiation, active filters, and oscillators.

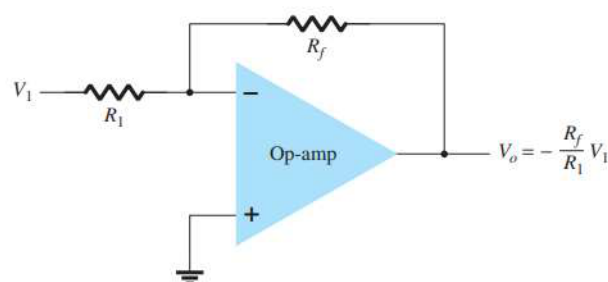


Figure 11: Inverting amplifier circuit.

Q5. Explain the non-inverting amplifier in detail, and explain the voltage follower. Compare it with the inverting amplifier.

Answer:

Non-inverting amplifier: the input is applied to the non-inverting (+) terminal, and the output is fed back to the inverting (-) terminal through R_f , with R_1 to ground. Because of the virtual short, $V_+ = V_- = V_{in}$.

$$V_{in}/R_1 = (V_o - V_{in})/R_f \Rightarrow A_{CL} = 1 + R_f/R_1$$

Characteristics: no phase inversion, gain = $1 + R_f/R_1$, very high input impedance, excellent signal fidelity.

Voltage follower: when $R_f = 0$ and R_1 is infinite, the gain becomes 1 and the circuit acts as a buffer, giving high input impedance and low output impedance without altering the signal.

Feature	Inverting	Non-Inverting
Input terminal	Inverting (–)	Non-inverting (+)
Output phase	180° (inverted)	In phase
Gain	$-R_f/R_1$	$1 + R_f/R_1$
Input impedance	Moderate	High

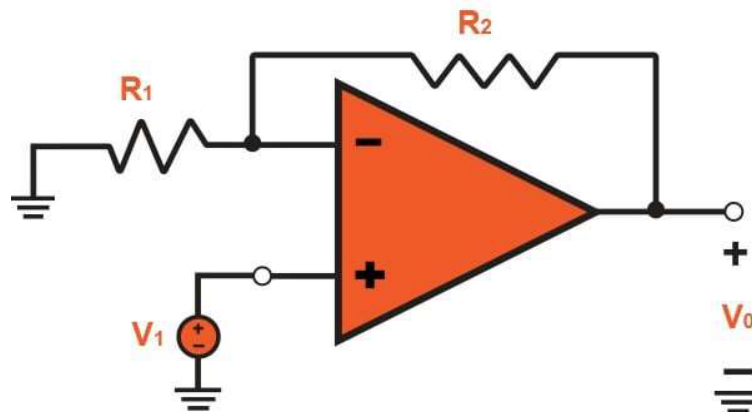


Figure 12: Non-inverting amplifier.

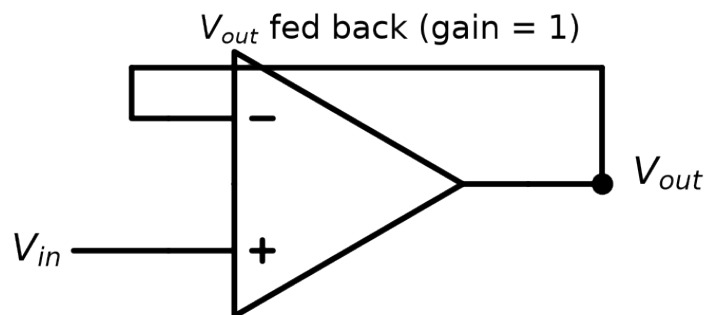


Figure 13: Voltage follower (buffer).

Q6. Explain the summing amplifier with a circuit diagram and derive its output. Also list the applications of op-amps.

Answer:

Summing amplifier: built on the inverting amplifier, it provides a means of algebraically summing several input voltages, each multiplied by a constant gain. Applying KCL at the virtual-ground node:

$$\begin{aligned} V_1/R_1 + V_2/R_2 + V_3/R_3 &= -V_o/R_f \\ V_o &= -R_f (V_1/R_1 + V_2/R_2 + V_3/R_3) \end{aligned}$$

If all resistors are equal, $V_o = -(V_1 + V_2 + V_3)$.

Applications of op-amps: signal amplification, active filters (low-pass, high-pass, band-pass), mathematical operations (integration, differentiation, addition, subtraction), voltage follower (buffer), oscillators and signal generators, and analog computation.

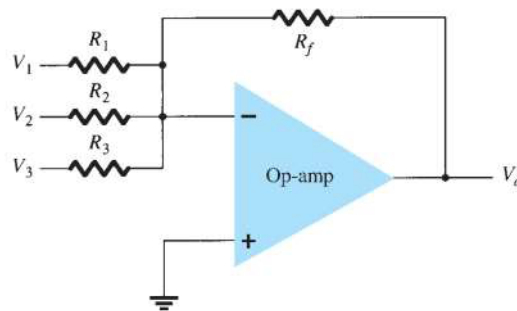


Figure 14: Three-input summing amplifier.

Q7. Explain the RC phase-shift oscillator in detail with circuit diagram, frequency of oscillation and gain condition.

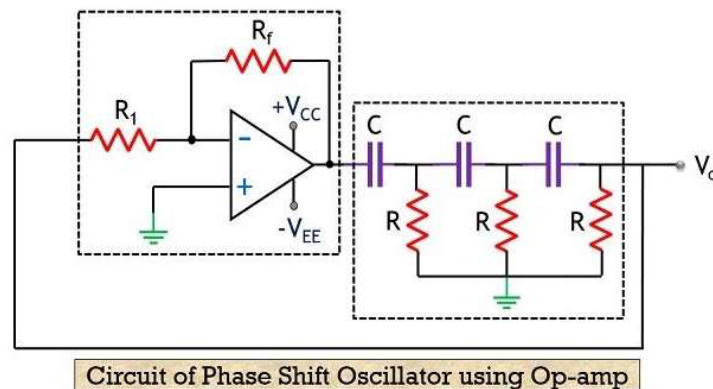
Answer:

A phase-shift oscillator is a sinusoidal oscillator that produces a stable sine wave using an amplifier and an RC phase-shift network. The RC network provides the phase shift required to make the feedback positive.

- The RC network has three identical sections, each providing 60° of phase shift, giving a total of 180° .
- The attenuation of the three-section network is $1/29$, so the amplifier gain must be greater than 29 for the loop gain βA to exceed unity.
- The op-amp gain is set by the resistors R_i and R_f .

Frequency of oscillation: $f = 1 / (2 \pi R C \sqrt{6})$

Feedback factor $\beta = 1/29$; condition: $\beta A > 1$, i.e. $A > 29$



Circuit of Phase Shift Oscillator using Op-amp

Figure 15: RC phase-shift oscillator.

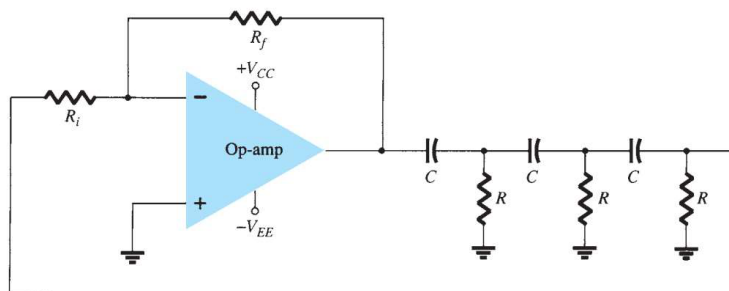


Figure 16: The three-section RC network and amplifier.

Q8. Explain the Wien bridge oscillator in detail with circuit diagram, frequency, gain condition, advantages and applications. Include a numerical example.

Answer:

A Wien bridge oscillator is a sinusoidal oscillator that uses an RC bridge network and an op-amp to generate low-distortion sine waves. It is widely used in audio frequency generators (20 Hz – 1 MHz).

- R_1, R_2, C_1, C_2 form the frequency-adjustment elements; R_3, R_4 form the feedback path.
- The op-amp output feeds the bridge at points a and c; the bridge output at b and d is the input to the op-amp.
- At the resonant frequency the bridge is balanced, giving zero phase shift and a feedback fraction of $1/3$.
- Using the Barkhausen criterion, the amplifier gain must be 3, i.e. $R_3/R_4 = 2$.

$$\text{Oscillation frequency: } f_0 = 1 / (2 \pi R C)$$

Numerical example: for $R_1 = R_2 = 22 \text{ k-ohm}$ and $C_1 = C_2 = 100 \text{ pF}$:

$$f_0 = 1 / (2 \pi \times 22 \times 10^3 \times 10^{-10}) = 72.34 \text{ kHz}$$

Advantages: high gain, no inductors (no magnetic interference), low distortion, changeable frequency, good frequency stability. **Applications:** audio signal generators, function generators, test equipment, communication and instrumentation circuits.

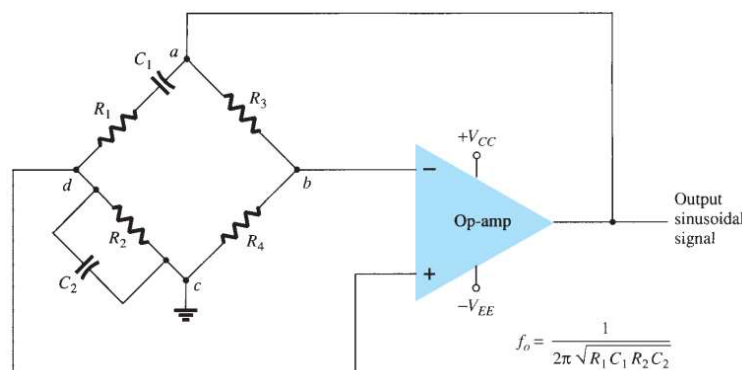


Figure 17: Wien bridge oscillator circuit.

$$f_0 = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 22 \times 10^3 \times 10^{-10}} = 72.34 \text{ kHz}$$

Figure 18: Numerical: frequency calculation.